Less Than Equal Constraints

3 constraints

$x_1 \le z_1$, $x_2 \le z_2$, $x_3 \le z_3$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \le \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

G x ≤ h

$x_1 \ge z_1$, $x_2 \ge z_2$, $x_3 \ge z_3$

$$\Rightarrow \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \le \begin{pmatrix} -z_1 \\ -z_2 \\ -z_3 \end{pmatrix}$$

Convert

$x_1 \ge z_1$, $x_2 \ge z_2$, $x_3 \ge z_3$

to

$-x_1 \le -z_1$, $-x_2 \le -z_2$, $-x_3 \le -z_3$

G x ≤ h

6 constraints , 3 variables

$y_1 \le x_1 \le z_1$, $y_2 \le x_2 \le z_2$, $y_3 \le x_3 \le z_3$,

$$\Rightarrow \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \le \begin{pmatrix} -y_1 \\ -y_2 \\ -y_3 \\ z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

6 x 3

G x ≤ h